

CS & IT ENGINEERING



Operating System

Process Synchronization

Lecture - 6

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Recap of Previous Lecture



Topic

Semaphore



Topics to be Covered



Topic

Questions in Semaphore

Topic

Classical Synchronization Problems

[NAT]



#Q. Consider a counting semaphore S , initialized with value 10. What should be the value of S after executing 6 times $P()$ and 8 times $V()$ function on S ?



#Q. Consider a semaphore S, initialized with value 37. Which of the following options gives the final value of $S=12$?

- A** Execution of 22 P() and 15 V()
- B** Execution of 25 P()
- C** Execution of 33 P() and 8 V()
- D** Execution of 31 P() and 6 V()

#Q. Consider a binary semaphore S, initialized with value 1. Consider 10 processes P1, P2 P10. All processes have same code as given below but, one process P10 has signal(S) in place of wait(S). If all processes to be executed only once, then maximum number of processes which can be in critical section together ?

```
process
{
    wait(S)
    C.S.
    signal(s)
}
```





Topic : Solution



P1, P2,, P9

process

```
{  
    wait(S)  
    C.S.  
    signal(s)  
}
```

P10

process

```
{  
    signal(S)  
    C.S.  
    signal(s)  
}
```

#Q. Consider a binary semaphore S, initialized with value 1. Consider 10 processes P1, P2 P10. All processes have same code as given below but, one process P10 has signal(S) in place of wait(S). If all processes to be executed only once, then maximum number of processes which can be in critical section together ?

```
while(True)
{
    wait(S)
    C.S.
    signal(s)
}
```





Topic : Solution



P1, P2,, P9

```
while(True)
{
    wait(S)
    C.S.
    signal(s)
}
```

P10

```
while(True)
{
    signal(S)
    C.S.
    signal(s)
}
```

[Question]



#Q. Consider a scenario where 2 processes are available: P0 and P1. The processes must be executed in order P0 then P1 then P0 then P1 and this sequence continues forever. Write a piece of code to control the sequence of execution with minimum number of semaphores?

$S_0 = 1, S_1 = 0$

binary

P0
wait(S_0)
=====
signal(S_1)

P1
wait(S_1)
=====
signal(S_0)

[Question]



#Q. Consider a scenario where 3 processes are available: P0, P1 and P2. The processes must be executed in order P1, P0 and P2 only. Write a piece of code to control the sequence of execution with minimum number of semaphores?

$S_0 = 0, S_1 = 1, S_2 = 0$

P0
wait(S_0)
=====
signal(S_2)

P1
wait(S_1)
=====
signal(S_0)

P2
wait(S_2)
=====
signal(S_1)

[MCQ]



#Q. Consider the following threads, T_1 , T_2 , and T_3 executing on a single processor, synchronized using three binary semaphore variables, S_1 , S_2 , and S_3 , operated upon using standard `wait()` and `signal()`. The threads can be context switched in any order and at any time. Which initialization of the semaphores would print the sequence BCABCABCA...?

T1	T2	T3
<pre>while(true) { wait (S₃); print("C"); signal(S₂); }</pre>	<pre>while(true){ wait(S₁); print("B"); signal(S₃);} </pre>	<pre>while(true) { wait(S₂); print("A"); signal(S₁); }</pre>

[2022]

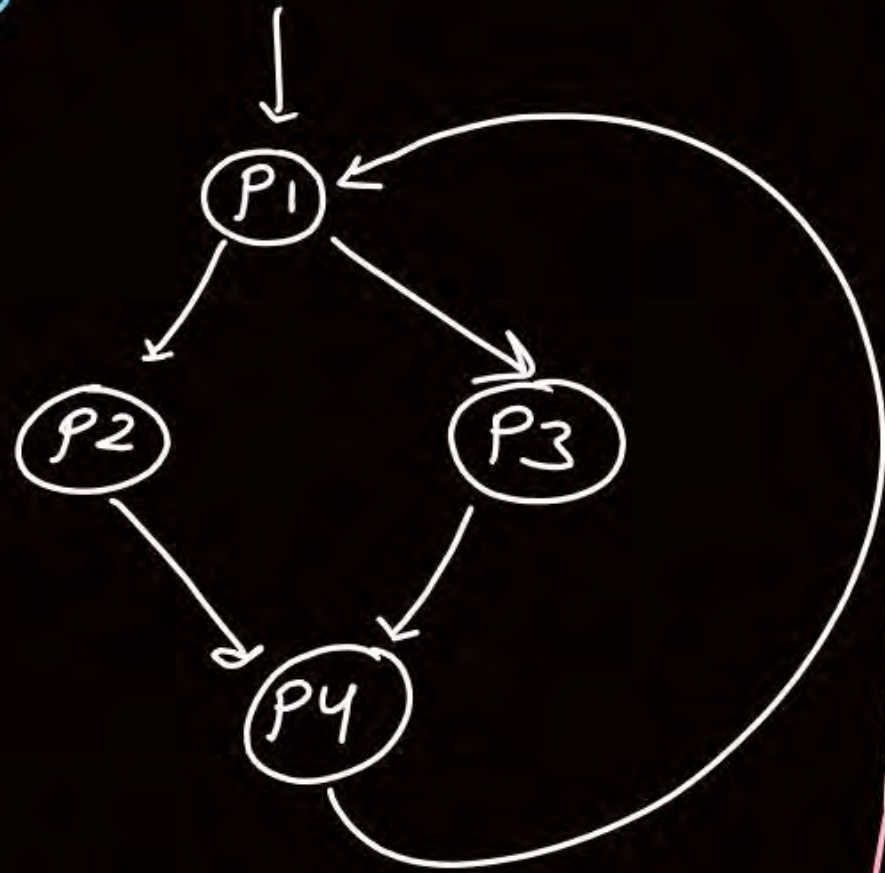
A $S_1 = 1; S_2 = 1; S_3 = 1$

B $S_1 = 1; S_2 = 1; S_3 = 0$

C ✓ $S_1 = 1; S_2 = 0; S_3 = 0$

D $S_1 = 0; S_2 = 1; S_3 = 1$

Ques)



$s_1 = 1$
 $s_2 = 0$
 $s_3 = 0$
 $s_{41} = 0$
 $s_{42} = 0$

P1
wait(s_1)
≡
signal(s_2)
signal(s_3)

P2
wait(s_2)
≡
signal(s_{41})

P3
wait(s_3)
≡
signal(s_{42})

P4
wait(s_{41})
wait(s_{42})
≡
signal(s_1)

#Q. Consider a non-negative counting semaphore S . The operation $P(S)$ decrements S , and $V(S)$ increments S . During an execution, 43 $P(S)$ operations and 27 $V(S)$ operations are issued in some order. The largest initial value of S for which at least 5 $P(S)$ operation will remain blocked is 11?

successful $P(S) = 43 - 5 = 38$

$$S + 27 - 38 = 0$$

$$S = 11$$

[NAT]



#Q. Consider a non-negative counting semaphore S . The operation $P(S)$ decrements S , and $V(S)$ increments S . During an execution, 20 $P(S)$ operations and 12 $V(S)$ operations are issued in some order. The largest initial value of S for which at least one $P(S)$ operation will remain blocked is 7.?

[2016]

successful $P(S) = 20 - 1 = 19$

$$S + 12 - 19 = 0$$

$$\boxed{S = 7}$$

[NAT]

Ans = 2



- #Q. A shared variable x , initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the process W and X reads x from memory, increments by one, stores it to memory and then terminates. Each of the processes Y and Z reads x from memory, decrements by two, stores it to memory and then terminates. Each processes before reading x invokes the P operation (i.e., wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What is the maximum possible value of x after all processes complete execution?

[2013]

$$x = 0$$

$\underline{W}$
 $P(S)$
 $R1 \leftarrow x$
 $R1 \leftarrow R1 + 1$
 $x \leftarrow R1$
 $V(S)$

$\underline{X}$
 $P(S)$
 $R2 \leftarrow x$
 $R2 \leftarrow R2 + 1$
 $x \leftarrow R2$
 $V(S)$

$\underline{Y}$
 $P(S)$
 $R3 \leftarrow x$
 $R3 \leftarrow R3 - 2$
 $x \leftarrow R3$
 $V(S)$

$\underline{Z}$
 $P(S)$
 $R4 \leftarrow x$
 $R4 \leftarrow R4 - 2$
 $x \leftarrow R4$
 $V(S)$

$$x = 0$$

$$S = 2$$

$\downarrow$
 race condition

$$0 + 1 + 1 - 2 - 2$$

$$x_{\min} = 0 - 2 - 2 = -4$$

All possible values of $x = 1, -2, 2, -4,$
 $-1, 0, -3,$

Ques) In prev Ques if S is initialized by 1. then

1. Max value of $x = -2$
2. min $\text{---} || \text{---} = -2$
3. All possible values of $x \Rightarrow \underline{\underline{-2}}$
single

↓
mutual exclusion
↓
no race condition

#Q. A shared variable x , initialized to zero, is operated on by four concurrent processes W, X, Y, Z as follows. Each of the process W and X reads x from memory, increments by 2, stores it to memory and then terminates. Each of the processes Y and Z reads x from memory, decrements by 3, stores it to memory and then terminates. Each processes before reading x invokes the P operation (i.e., wait) on a counting semaphore S and invokes the V operation (i.e., signal) on the semaphore S after storing x to memory. Semaphore S is initialized to two. What are the total distinct possible values of x after all processes complete execution?

Race
Conditⁿ

W	X	Y	Z
$x = x + 2$	$x = x + 2$	$x = x - 3$	$x = x - 3$

$$0 + 2 + 2 - 3 - 3$$

$$2, -3, 4, -6, -1$$

$$1, -4, -2$$

[MCQ]



#Q. Consider the two functions incr and decr shown below.

```
incr() {  
    wait(s);  
    X = X+1;  
    signal(s);  
}
```

```
decr() {  
    wait(s);  
    X = X-1;  
    signal(s);  
}
```

*5 Times $\Rightarrow$ incr()
3 Times $\Rightarrow$ decr()
 $X = 10$*

There are 5 threads each invoking incr once, and 3 threads each invoking decr once, on the same shared variable X. The initial value of X is 10.

Suppose there are two implementations of the semaphore s, as follows:

I1: s is a binary semaphore initialized to 1.

I2: s is a counting semaphore initialized to 2. *$\rightarrow$ race condition*

Let V1, V2 be the values of X at the end of execution of all the threads with implementations I1, I2, respectively.

Which one of the following choices corresponds to the minimum possible values of V1, V2, respectively? **[2023]**



15, 7



7, 7



✓ 12, 7



12, 8

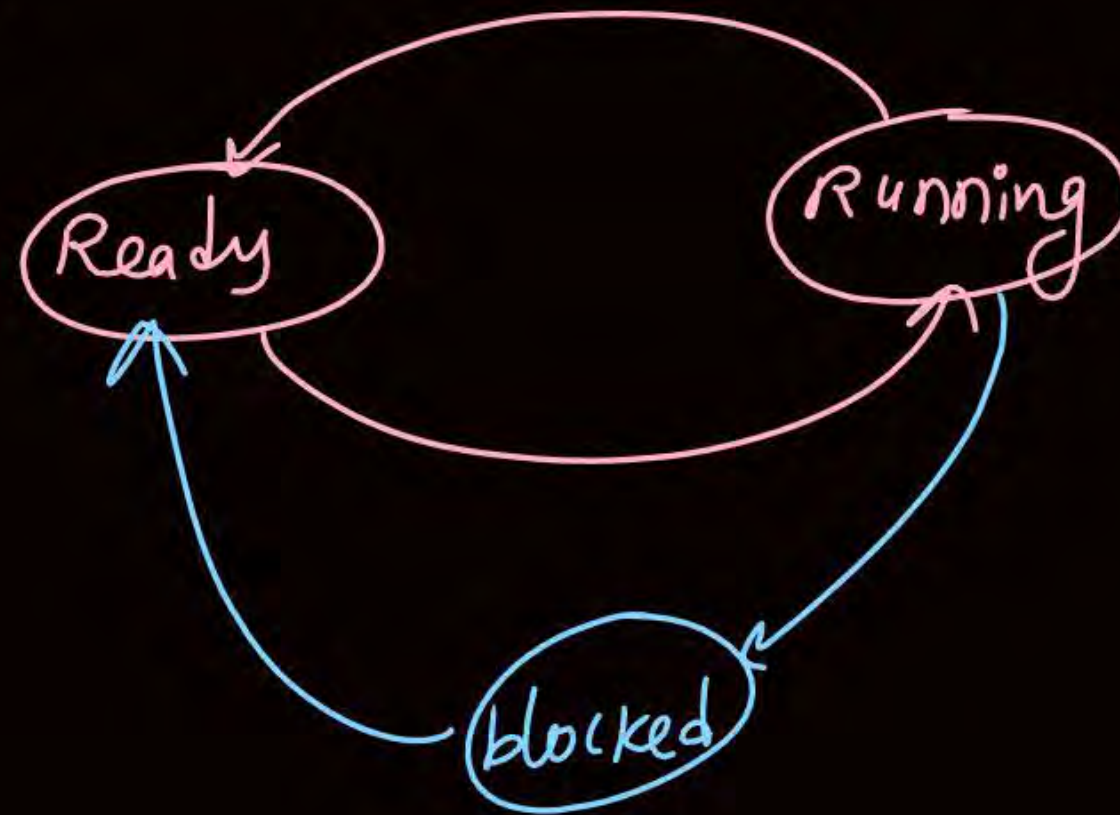
$s = 0$

wait(s)

```
{  
→ while (s <= 0);  
  s--;  
}
```

Busy waiting (Spin lock)

Process runs on CPU but waits for C.S.





Topic : Solutions Without Busy Waiting

```
wait(Semaphore s){
```

```
    s=s-1;
```

```
    if (s<0) {
```

```
        // add process to queue
```

```
        block();
```

```
    }
```

```
}
```

$p_1 \Rightarrow \cancel{C.S.}$

$p_2 \Rightarrow \cancel{C.S.}$

$p_3 \Rightarrow C.S.$

$s = \cancel{1} \phi$

$\cancel{= 1}$

$\cancel{= 2}$

$\cancel{= 1}$

$\phi - 1$



```
signal(Semaphore s){
```

```
    s=s+1;
```

```
    if (s<=0) {
```

```
        // remove
```

```
        process p from queue
```

```
        wakeup(p);
```

```
    }
```

```
}
```

$\cancel{wait(s)}$

$\cancel{C.S.}$

$\cancel{signal(s)}$

#Q. Given below is a program which when executed spawns two concurrent processes:

Semaphore X:=0;

/* Process now forks into concurrent processes P1 & P2 */

P1 : repeat forever

P2:repeat forever

V(X);

P(X);

Compute;

Compute;

P(X);

V(X);

Consider the following statements about processes P1 and P2:

I: It is possible for process P1 to starve.

II. It is possible for process P2 to starve.



2 mins Summary

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Questions in Semaphore

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Classical Synchronization Problems





Happy Learning

THANK - YOU

